Proof of Arkadeep Dutta's edge-collision ratio conjecture (OEIS A396659)

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Abstract

We prove the conjecture in Arkadeep Dutta's OEIS A396659, revision 16, FORMULA, third statement [1]: $T(n, n^2)/T(n, n^2+1) = (2n-1)/n$ for every $n \geq 2$. Here $T(n, k)$ counts length-k walks from a fixed vertex of the complete looped directed graph on n labeled vertices, visiting every vertex and repeating an edge for the first time at step k. An omitted-edge classification and marked loop insertion prove the ratio. We also establish the entry's two companion formulas, $T(n, n+1) = n!$ and $T(n, n^2+1) = (n!)^n$, with the linear-walk normalization explicit.

1 The counting convention and result

The conjecture in A396659 is resolved by counting the two possible types of almost-complete trail. We give that argument and prove both other statements in the same FORMULA field [1]. Fix a set V of $n \geq 1$ vertices and a starting vertex $v \in V$. The graph has exactly one directed edge $a \to b$ for every ordered pair $(a, b) \in V^2$, including one loop (a, a) at each vertex. A length-k walk is a vertex list $(x_0, \ldots, x_k)$ with $x_0 = v$; its edges are (x_{i-1}, x_i) for $1 \leq i \leq k$. We count the walk in $T(n, k)$ when its first $k-1$ edges are distinct, its last edge repeats an earlier edge, and its vertices include all of V. The list has a distinguished first position: there is no quotient by cyclic rotation, graph automorphisms, or reversal. These are the conventions of A396659, revision 16 [1].

Let $E_n(v)$ be the number of length-n^2 walks from v that use every edge exactly once. Such a walk is an Eulerian circuit, represented linearly with specified initial vertex v. The following identities give all three source formulas.

Theorem 1. *For every $n \geq 2$ and $v \in V$, the integer $E_n(v)$ is positive and*

$$T(n, n^2+1) = nE_n(v), \tag{1}$$

$$T(n, n^2) = (2n-1)E_n(v). \tag{2}$$

Consequently $T(n, n^2+1) > 0$ and

$$\frac{T(n, n^2)}{T(n, n^2+1)} = \frac{2n-1}{n}.$$

Moreover, for every $n \geq 1$,

$$T(n, n+1) = n!, \qquad T(n, n^2+1) = (n!)^n. \tag{3}$$

The pinned source `CollisionPrefix.lean` [4] develops the ratio proof and its loop-insertion bijection. For the companion factorial formula we use the classical last-exit-tree construction of the BEST theorem [2, §6, Theorems 5a–6], writing out the count for a fixed starting vertex rather than a fixed first edge or a cyclic tour. The tree count is the Prüfer enumeration [3, Chapter 9, appendix, p. 180].

2 Almost-complete trails

Assume $n \geq 2$ in this section and the next. A trail is a walk with no repeated edge. For a trail P from v to t, let $d_P^+(x)$ and $d_P^-(x)$ count its outgoing and incoming edges at x. Telescoping along the vertex list gives

$$d_P^+(x) - d_P^-(x) = \mathbf{1}_{x=v} - \mathbf{1}_{x=t}. \tag{4}$$

Loops contribute once to each degree and therefore cancel in this identity. In particular every walk counted by $E_n(v)$ is closed, since the full graph has equal incoming and outgoing degree n at every vertex.

Lemma 1. *A length-(n^2-1) trail from v visits every vertex. If its missing edge is (a,b) and its endpoint is t, exactly one of the following holds:*

1. *$b = v$ and $a = t$;*
2. *$a = b \neq v$ and $t = v$.*

Proof. There are n^2 edges, so there is a unique missing edge (a,b). For every $x \in V$, choose $y \neq b$, possible because $n \geq 2$. The edge (x,y) is present, so x occurs in the trail. Removing (a,b) from the full graph and using (4) gives

$$\mathbf{1}_{x=v} - \mathbf{1}_{x=t} = \mathbf{1}_{x=b} - \mathbf{1}_{x=a} \qquad (x \in V). \tag{5}$$

If $a \neq b$, the right side has its unique positive entry at b and negative entry at a; hence $b = v$ and $t = a$. If $a = b$, the right side is zero, so $t = v$. The subcase $a = b = v$ belongs to case 1, while $a = b \neq v$ is case 2. The cases are disjoint. □

3 Two bijections

Call a trail in case 1 a return trail. Appending its missing edge (t,v) produces an Eulerian circuit. Conversely, delete the final edge of an Eulerian circuit: closedness ensures that the missing edge is (t,v). These operations preserve the fixed start and are inverses. There are therefore exactly $E_n(v)$ return trails.

For $a \neq v$, let L_a be the number of length-(n^2-1) trails from v whose missing edge is the loop (a,a). They are closed by Lemma 1. In any one of these trails there are exactly $n-1$ linear positions at which to insert (a,a): the positions whose current vertex is a, immediately before the next edge. Indeed, there are $n-1$ used outgoing edges at a, and neither endpoint is a, so each occurrence of a is the tail of exactly one such edge.

Inserting the loop at a marked position produces an Eulerian circuit. Conversely, an Eulerian circuit contains (a,a) exactly once; deleting that edge, or equivalently its repeated head vertex, gives a trail with a marked insertion position at a. These constructions are inverse even when the vertex a occurs many times: the position, not just the vertex, is retained. Thus

$$(n-1)L_a = E_n(v) \qquad (a \neq v), \qquad \sum_{a \neq v} L_a = E_n(v). \tag{6}$$

The second identity follows by summing the first over the $n-1$ possible vertices and cancelling the positive integer $n-1$. The condition $a \neq v$ is essential: at v both endpoints are available as distinct linear cuts.

4 The two collision counts

Deleting the last edge of a walk counted by $T(n, n^2+1)$ leaves an Eulerian circuit. Conversely, any of the n outgoing edges at the endpoint v of such an Euler walk repeats an edge. Coverage is automatic because all edges have already occurred. This proves (1).

Deleting the last edge of a walk counted by $T(n, n^2)$ leaves a trail of length $n^2 - 1$, hence one of the disjoint classes in Lemma 1. For a return trail ending at t, the edge (t, v) is missing and the other $n-1$ outgoing edges repeat. For a trail missing (a, a) with $a \neq v$, the endpoint is v, and all n outgoing edges at v repeat. All these prefixes already visit every vertex. Deleting and appending the last edge are inverse operations, so (6) yields

$$T(n, n^2) = (n-1)E_n(v) + n\sum_{a \neq v} L_a = (2n-1)E_n(v).$$

5 The companion formulas and positivity

We now count $E_n(v)$, which will also justify cancellation in the ratio. For $n \geq 2$, select the last outgoing edge $x \to p(x)$ of a full Euler walk at each $x \neq v$. These edges form a tree directed toward v. Indeed, after that edge reaches $p(x) \neq v$, the walk must leave $p(x)$ later, since it ends at v. Last-exit times therefore strictly increase along parent edges not reaching v; a directed cycle is impossible. Every parent chain must consequently reach v.

Conversely, choose such a tree and order the outgoing edges at every vertex, putting its tree edge last whenever $x \neq v$. Follow the next unused outgoing edge from v until stuck. It can stop only at v: otherwise (4) would require $n+1$ used incoming edges at the endpoint whose n outgoing edges were exhausted. The unused edges are balanced. Let S be the vertices with unused outgoing edges; then $v \notin S$. If $x \in S$, its last, tree edge remains unused. Its parent has unused indegree, hence unused outdegree, so lies in S. Following parents to v gives a contradiction. Thus all edges were used. The walk recovers its local orders and tree, so the constructions are inverse.

For clarity, the labeled-tree count is n^{n-2}: repeatedly delete the smallest leaf and record its neighbor until two labels remain. This gives a word of length $n-2$ over V. Conversely, assign each label degree one plus its multiplicity in the word. Join the smallest label of degree one to the next letter, delete that label, and decrement the letter's degree; finally join the two remaining labels. At each step a degree-one label absent from the remaining word exists, and these operations are inverse. This is the smallest-label variant of the Prüfer correspondence [3]; when $n = 2$ the word is empty and there is one tree. Orienting toward v is unique.

For each tree there are $n!$ local orders at v and $(n-1)!$ at every other vertex. Hence

$$E_n(v) = n^{n-2}n!\big((n-1)!\big)^{n-1} = n^{n-1}\big((n-1)!\big)^n > 0. \tag{7}$$

Together with (1) and (2), this proves the ratio and $T(n, n^2+1) = (n!)^n$. For $n = 1$ the only edge is the loop, so $E_1(v) = T(1, 2) = 1$ directly.

Finally, consider a walk $(x_0, \ldots, x_{n+1})$ counted by $T(n, n+1)$. Its final repeated edge adds no new vertex, so the prefix $(x_0, \ldots, x_n)$ visits all n vertices. It has exactly one repeated vertex occurrence. Since x_n is the tail of an earlier edge, that occurrence must be x_n. Thus $x_0, \ldots, x_{n-1}$ is a permutation of V beginning at v; x_n is any of the n vertices. Its unique earlier occurrence determines x_{n+1}. Conversely each such choice works: the first n edges have distinct tails and the last edge repeats the earlier outgoing edge from x_n. There are $(n-1)!n = n!$ choices, also for $n = 1$. This completes (3) and Theorem 1.

6 Finite checks and verification scope

A separate standard-library Python checker counts trails by used-edge subsets and endpoint, appending a repeated edge only when vertex coverage holds. It reproduces the complete OEIS rows for $1 \le n \le 4$. Explicit enumeration for $n = 2, 3$ additionally checks the omitted-edge partition, the return-trail bijection, and both directions of marked loop insertion, together with the last-exit-tree correspondence and its inverse. The last two counts are:

n	$E_n(v)$	$T(n, n^2)$	$T(n, n^2 + 1)$
2	2	6	4
3	72	360	216
4	82 944	580 608	331 776

These are finite diagnostic checks. The written proofs establish all three formulas for their stated ranges. The pinned formal source [4] contains the ratio and positivity declarations; the companion-formula proofs here are not claimed as Lean-verified results.

Availability. The source package contains this note, the Python 3 standard-library checker `verify.py`, its exact output, and rebuilding instructions. The proofs have no computational dependency.

References

[1] Arkadeep Dutta, *A396659*, The On-Line Encyclopedia of Integer Sequences, revision 16, 29 June 2026; author field dated 1 June 2026. `https://oeis.org/A396659/internal` (accessed 1 November 2026).

[2] T. van Aardenne-Ehrenfest and N. G. de Bruijn, *Circuits and trees in oriented linear graphs*, Simon Stevin **28** (1951), 203–217, §6. `https://pure.tue.nl/ws/files/3311129/597493.pdf`. Reprinted in *Classic Papers in Combinatorics* (2009), 149–163, `https://doi.org/10.1007/978-0-8176-4842-8_12`.

[3] R. P. Stanley, *Topics in Algebraic Combinatorics*, version of 1 February 2013. Chapter 9, appendix, p. 180 (Prüfer codes); Chapter 10, Theorem 10.2, pp. 191–192 (last exits). `https://math.mit.edu/~rstan/algcomb/algcomb.pdf`.

[4] *CollisionPrefix.lean*, the-omega-institute/trureturing, revision `d1876cfbbef8c13023311126691b180eefa69569`, declarations `collision_ratio` and `long_count_positive`. Commit-pinned source.